Lab 3# - GAIT Analysis

BME/CS 479 - Wearables and Nearables Technology Laboratory – Spring 2026 Rafael Mejia | Group 7

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Hardware Design and Development

In this project we used the FireBeetle 328P which was the brain, transmitting data via BLE to Processing UI. Adafruit MPU-6050 which is the accelerometer on the ankle. The FSRs were attached to the footsole. We also had LEDs to detect pressure on each FSR. The design is made to be worn on the ankle to keep it small and clean. Some assembly challenges were creating a design to keep the LEDs visible and then wiring all the components together.

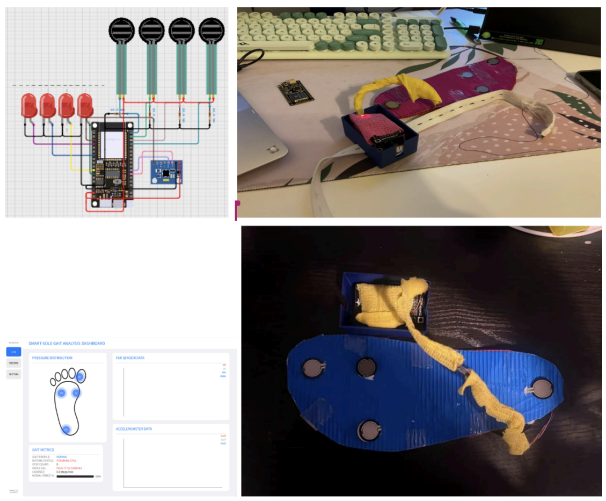

Figure: The image shows the assembled device (2 inches wide × 2 inches high) and also includes the circuit diagram and UI screenshot.

User Interface Design and Development

The user interface was made simple yet informative. There are 3 modes, the first being a live mode where the user can see live metrics and pressure points. The second is the record mode that shows live metrics and history as well. The final mode is the Guitar hero mode which is a rhythm game where the user must step to input. One thing we can improve is the special feature of having actual songs to play along with to make it more fun. We can also improve the insole design to fit in the shoe better.